SLIIT Department of Software Engineering - Faculty of Computing

Mobile Application Project: Farmora

Milestone 1: UX Team Formation, User Research & Key Flows (LO1, LO2)

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1. Project Title & Overview

Project Name: Farmora ? AI-Powered Smart Agricultural Ecosystem & Traceable Supply Chain
Domain: AgriTech / Supply Chain & Digital Marketplaces
Target Region: Sri Lanka (Western, Central, North-Western & Northern Provinces)
App Type: Cross-Platform Mobile & Progressive Web Application (Flutter + Firebase Serverless)

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2. UX Team Formation & Roles (Lab 01 & Lab 02)

To satisfy the semester lab requirements, team members took ownership of distinct user experiences:

Member Name	Student ID	Designated UX Role	Focus Modules & Responsibiliti
Ahsan Mohammed	ITXXXXXXX (Lead)	Lead Architect & Admin/Buyer U	Overall system architecture, B
Kajana S.	ITXXXXXXX	Farmer Operations & Agronomy R	Farmer registration, Harvest l
Sureka (Suka)	ITXXXXXXX	Logistics & Transporter UX	Transporter dispatching, Fleet
Team Member 4	ITXXXXXXX	UX Research & QA Lead	Usability benchmarking, Access

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3. User Groups & Persona Identification (Lab 03)

Persona 1: Sunimal Perera (Smallholder Farmer - Dambulla)

- Age:** 48
- Context:** Operates a 2-acre vegetable farm; experiences 25-35% post-harvest spoilage and price exploitation by middlemen at economic centres.
- Digital Literacy:** Low-to-Medium (WhatsApp, YouTube, voice notes; struggles with complex text menus).
- Core Goals:** Fair market pricing without broker cartels; rapid harvest listing; agronomic disease detection without costly consultants.
- App Needs:** Simple 1-tap harvest logging, voice search, high-contrast visual cues, Sinhala/Tamil support, QR code traceability stickers.

Persona 2: Sarah Mendis (Supermarket/Hotel Food Procurement Manager - Colombo)

- Age:** 34
- Context:** Procures 1,500kg of fresh vegetables daily for retail distribution; requires grade-A consistency, timely delivery, and farm-to-shelf

traceability.
- Digital Literacy:** High (Enterprise ERPs, mobile banking, inventory apps).
- Core Goals:** Verified quality grading, transparent cold-chain verification, escrow-protected payments, automated VAT invoicing.
- App Needs:** Bulk ordering, real-time dispatch tracking, digital certificates, escrow release on quality verification.

Persona 3: Ruwan Bandara (Independent Fleet Transporter - Kurunegala)

- Age:** 41
- Context:** Owns 2 refrigerated trucks (Reefers) and 1 light lorry; frequently returns empty on return trips (deadhead miles).
- Digital Literacy:** Medium (Google Maps, phone calls, SMS).
- Core Goals:** Optimize load capacity, minimize empty backhauls, monitor reefer temperature to prevent cargo rejection.
- App Needs:** Real-time cargo matching, clear turn-by-turn routing, instant digital Proof-of-Delivery (photo + signature).

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4. User Research Methodology & Findings (Lab 04)

Research Methodology:

- Semi-structured Interviews:** 12 participants (5 Farmers in Dambulla/Nuwara Eliya, 4 Buyers/Wholesalers in Colombo/Pettah, 3 Transporters).
- Contextual Inquiry:** Observed harvest packing and middleman negotiation at the Dambulla Dedicated Economic Centre.
- Survey:** Distributed to 35 agricultural cooperative members via mobile forms.

Key Pain Points Discovered:

1. **Opaque Pricing:** 83% of farmers reported receiving up to 40% less than Colombo retail price due to multi-tiered intermediaries.
2. **Cold-Chain Spoilage:** Over 28% of perishable produce degrades due to lack of temperature logging during transit.
3. **Payment Defaults:** Buyers and farmers experience trust deficits; farmers fear delayed payments, buyers fear substandard rot.
4. **Language Barrier:** Most agricultural apps are English-only, completely alienating local farmers.

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5. Validated Key User Flows (Lab 05)

Flow 1: Farmer Harvest Onboarding & Video Quality QR Generation

‘[Farmer Log In]‘ ? ‘[Capture Harvest Photos/Video]‘ ? ‘[Auto-detect Grade & Price Guide]‘ ? ‘[Publish Batch]‘ ? ‘[Generate Batch QR Sticker]‘ ? ‘[Awaiting Buyer Purchase]‘

Flow 2: Buyer Direct Purchase & Escrow Settlement

‘[Browse Verified Marketplace]‘ ? ‘[Inspect Video/Batch Audit]‘ ? ‘[Add to Cart & Select Delivery Option]‘ ? ‘[Deposit to PayHere/Escrow]‘ ? ‘[Real-Time Tracking]‘ ? ‘[Inspect at Delivery & Scan QR]‘ ? ‘[Release Escrow Funds to Farmer]‘

Flow 3: Transporter Fleet Allocation & Delivery POD

‘[Dispatch Notification Received]‘ ? ‘[Select Vehicle & Driver]‘ ? ‘[Accept Route with Cold-Storage Parameters]‘ ? ‘[GPS Live Tracking]‘ ? ‘[Upload Delivery Photo & Recipient Signature]‘ ? ‘[Instant Payout Credited]‘

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6. Sketching & Low-Fidelity Wireframes (Lab 06)

Low-fidelity sketches and information hierarchy were designed covering:
1. **Adaptive Role-Based Dashboard:** Tailored navigation for Farmer, Buyer, Transporter, and Agri-Admin.
2. **Accessible Form Layouts:** Large touch targets (min 48dp), clear step progression, and voice input integration.
3. **Marketplace Matrix:** Card-based browse layout displaying grade, harvest timestamp, location, and verified badges.
4. **Fleet Telematics View:** Combined map and cold-storage temperature telemetry widget.